Convergence Tests

Section 1: Integral Test and p-Series

1 For what values of p does the p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converge?

a. $p > 1$

b. $p \geq 1$

c. $p < 1$

d. $p > 0$

2 Use the Integral Test to determine the convergence of $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$.

a. Diverges

b. Converges to 1

c. Converges to $\ln 2$

d. Inconclusive

3 Which conditions must $f(x)$ satisfy to apply the Integral Test to $\sum a_n$ where $a_n = f(n)$?

a. f must be continuous, positive, and decreasing on $[1, \infty)$

b. f must be differentiable and alternating

c. f must be bounded and periodic

d. $f'(x) > 0$ everywhere

Section 2: Comparison Tests (Direct and Limit)

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4 Determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^3+5}$.

a. Converges by Direct Comparison with $\sum \frac{1}{n^3}$

b. Diverges by comparison with $1/n$

c. Inconclusive

d. Diverges by the Divergence Test

5 Use the Limit Comparison Test with $b_n = 1/n$ to test $\sum_{n=1}^{\infty} \frac{2n+1}{n^2-3}$.

a. Diverges because $L = 2 > 0$ and $\sum 1/n$ diverges

b. Converges because $L = 0$

c. Converges because $L = 2$

d. Inconclusive

6 Which comparison series b_n is best suited for testing $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+4n}$ via LCT?

a. $b_n = \frac{1}{n^{3/2}}$

b. $b_n = \frac{1}{n^2}$

c. $b_n = \frac{1}{n}$

d. $b_n = \frac{1}{\sqrt{n}}$

Section 3: Alternating Series and Error Bounds

7 What are the two conditions of the Alternating Series Test (AST) for $\sum (-1)^n b_n$ ($b_n > 0$)?

a. $b_{n+1} \le b_n$ for all n and $\lim_{n\to\infty} b_n = 0$

b. $\sum b_n$ converges and $b_n > 0$

c. $\lim_{n\to\infty} b_n = 1$ and b_n is decreasing

d. b_n is geometric

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8 Classify $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$.

a. Conditionally convergent

b. Absolutely convergent

c. Divergent

d. Oscillating divergent

9 If $S = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^4}$ is approximated by S_3, what is the maximum error by the Alternating Series Estimation Theorem?

a. $|R_3| \le \frac{1}{4^4} = \frac{1}{256}$

b. $|R_3| \le \frac{1}{3^4} = \frac{1}{81}$

c. $|R_3| \le \frac{1}{5^4}$

d. $|R_3| \le \frac{1}{16}$

Section 4: Ratio and Root Tests

10 State the conclusion of the Ratio Test when $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$.

a. The series converges absolutely

b. The series converges conditionally

c. The series diverges

d. The test is inconclusive

11 Apply the Ratio Test to $\sum_{n=1}^{\infty} \frac{3^n}{n!}$.

a. $L = 0 < 1 \Rightarrow$ Converges absolutely

b. $L = 3 > 1 \Rightarrow$ Diverges

c. $L = 1 \Rightarrow$ Inconclusive

d. $L = \infty \Rightarrow$ Diverges

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12 Use the Root Test to determine the convergence of $\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-1}\right)^n$.

a. $L = 2/3 < 1 \Rightarrow$ Converges absolutely

b. $L = 2/3 \Rightarrow$ Diverges

c. $L = 1 \Rightarrow$ Inconclusive

d. $L = 0$

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